

A Minimalist Putback-Based Bidirectional Programming Language, Formally Verified

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1 Introduction

Theory construction in terms of AGDA programming

2 Decidable partiality

The bidirectional transformations discussed in this paper are “decidably partial”. By “decidable partiality” we mean that the transformations are actually total functions which wrap their results in the standard **Maybe** datatype:

```
data Maybe (A : Set) : Set where
  just    : A → Maybe A
  nothing : Maybe A
```

The result of a piece of successful computation is wrapped in the **just** constructor, while failed computation is represented by **nothing**. The behaviour of decidably partial functions is different from the usual partial functions as formalised in terms of complete partial orders (see, e.g., Gibbons [4]): In Section 4.3 we will need to define programs that produce some result or fail respectively when a sub-program fails or produces some result, but this is not possible in the setting of complete partial orders (since such functions violate monotonicity, a necessary condition for computability).

We can choose to write our decidably partial bidirectional transformations directly as **Maybe**-programs, but when proving their well-behavedness, we will need to analyse or establish how a piece of decidably partial computation succeeds or fails, and the task would be much easier if the computation steps leading to success or failure are reified as data. (An example of such proofs is given in the definition of lens composition $_ \leftrightarrow _$ in Section 3.) The reification starts from the following datatype **Par**, which is a deep embedding of the combinators that we will use for writing decidably partial programs:

```
data Par : Set → Set1 where
  return      : { A : Set } → A → Par A
```

```

runPar : { A : Set } → Par A → Maybe A
runPar (return x) = just x
runPar (mx ≫= f) with runPar mx
runPar (mx ≫= f) | just x  = runPar (f x)
runPar (mx ≫= f) | nothing = nothing
runPar fail = nothing
runPar (catch mx f my) with runPar mx
runPar (catch mx f my) | just x  = runPar (f x)
runPar (catch mx f my) | nothing = runPar my
runPar (assert true then mx) = runPar mx
runPar (assert false then mx) = nothing

```

Fig. 1. Definition of the interpreter of Par to Maybe

```

_≫=_      : { A B : Set } → Par A → (A → Par B) → Par B
fail      : { A : Set } → Par A
catch     : { A B : Set } → Par A → (A → Par B) → Par B → Par B
assert_then_ : { A B : Set } → Bool → Par A → Par A

```

Informally: A value of type `Par A` represents a piece of decidable partial computation that either successfully produces a value of type `A` or fails. The first two constructors `return` and `_≫=_` represent the usual monadic operators [5,8], and `fail` indicates failed computation. We will also need error handling, which is represented by the `catch` constructor. The behaviour of our `catch` is slightly different from the usual one: execution of `catch mx f my` binds the result of `mx` to `f` if `mx` computes successfully, or goes to execution of `my` if `mx` fails to compute. Also we will frequently need to ensure that a `Bool` value is `true` before continuing with the rest of a program, so the constructor `assert_then_` is included for convenience — execution of `assert b then p` becomes execution of `p` when `b` is `true`, or fails otherwise.

Formally, the meaning of the constructors of `Par` is given by the interpreter $runPar : \{A : \text{Set}\} \rightarrow \text{Par } A \rightarrow \text{Maybe } A$ shown in Figure 1. We say that $mx : \text{Par } A$ computes to $x : A$ exactly when $runPar \ mx \equiv \text{just } x$, and that mx fails to compute exactly when $runPar \ mx \equiv \text{nothing}$. The first equality is what we aim to reify as data next.

Our reification of decidable partial computation as the datatype `Par` and AGDA’s support of full dependent types allow us to compute for any $mx : \text{Par } A$ and $x : A$ a type $mx \mapsto x$ whose inhabitants explain how mx computes to x step by step. Below is a sample of the definition of $_ \mapsto _$:

```

_↦_ : { A : Set } → Par A → A → Set1
return x ↦ x' = x ≡ x'
mx ≫= f ↦ y = (x : _) × (mx ↦ x) × (f x ↦ y)
...

```

That is, `return x` computes to x' when x is exactly x' , and $mx \gg= f$ computes to y when there exists $x : A$ such that mx computes to x and $f \ x$ computes to y

(and so on). Here our use of the term “computes to” is justified, since we can prove that

$$mx \mapsto x \quad \text{and} \quad \text{runPar } mx \equiv \text{just } x$$

are equivalent. That is, a term of type $mx \mapsto x$ can be constructed if and only if $\text{runPar } mx \equiv \text{just } x$, meaning that our reification of $\text{runPar } mx \equiv \text{just } x$ as $mx \mapsto x$ is accurate. We are thus entitled to use $\mapsto$ for stating various properties involving successful decidable partial computation, in particular well-behavedness of bidirectional transformations in the next section.

3 Bidirectional transformations formalised

We follow the classic (well-behaved) lens approach [2] to bidirectional transformations, but adopting the variation used by Pacheco et al [6]:

Definition 1 (lenses). *A (well-behaved) lens between a source type $S : \text{Set}$ and a view type $V : \text{Set}$ is a pair of functions*

$$\text{put} : S \rightarrow V \rightarrow \text{Par } S \quad \text{and} \quad \text{get} : S \rightarrow \text{Par } V$$

satisfying the PutGet law:

$$\text{put } s \ v \mapsto s' \quad \text{implies} \quad \text{get } s' \mapsto v \quad \text{for all } s, s' : S \text{ and } v : V$$

and the GetPut law:

$$\text{get } s \mapsto v \quad \text{implies} \quad \text{put } s \ v \mapsto s \quad \text{for all } s : S \text{ and } v : V.$$

In AGDA the above becomes just a record definition:

```
record Lens (S V : Set) : Set1 where
  field
    put : S → V → Par S
    get : S → Par V
    PutGet : {s s' : S} {v : V} → (put s v ↦ s') → (get s' ↦ v)
    GetPut : {s : S} {v : V} → (get s ↦ v) → (put s v ↦ s)
```

Informally: Execution of $\text{put } s \ v$ should produce an updated version of the source s by properly embedding all information of the view v into s . The *PutGet* law ensures that put does its job properly by requiring that, after put updates a source with a view, get can completely recover the view from the updated source. The *GetPut* law, on the other hand, ensures that put does nothing more than its designated job by requiring that put performs no update on a source if the view it receives is directly extracted from the source by get . Note that our definition of lenses is stronger than the one given by Foster et al [2]: our *PutGet* law says that successful computation of put guarantees success of the

subsequent computation of *get*, whereas in Foster et al’s definition there is no such guarantee; the *GetPut* law is similar.

A related notion is “partial” isomorphisms as defined by

```

record Iso (A B : Set) : Set1 where
  field
    to    : A → Par B
    from  : B → Par A
    to-from-inverse : {x : A} {y : B} → (to x ↦ y) → (from y ↦ x)
    from-to-inverse : {x : A} {y : B} → (from y ↦ x) → (to x ↦ y)

```

Lens is a generalisation of *Iso* because we can easily convert *Iso* *A B* to *Lens* *A B*, whose *put* directly produces a new source from its view argument, ignoring its source argument:

```

iso-lens : {A B : Set} → Iso A B → Lens A B
iso-lens iso = record
  { put = λ _ b → Iso.from iso b
  ; get = Iso.to iso
  ; PutGet = Iso.from-to-inverse iso
  ; GetPut = Iso.to-from-inverse iso }

```

This conversion gives us a first way to manufacture lenses. For example, the identity isomorphism between the same type — whose *to* and *from* are both *return* — gives rise to the “replacing” lens, and the empty isomorphism between any two types — whose *to* and *from* are both $\lambda _ \rightarrow \text{fail}$ — gives rise to the “failure” lens. Both lenses will appear in Section 4.1.

To give the reader a taste of how well-behavedness proofs are constructed in our AGDA setting, let us try to define an operator $_ \leftrightarrow _$ for lens composition:

```

_↔_ : {A B C : Set} → Lens A B → Lens B C → Lens A C
l ↔ r = record
  { put = λ a c → Lens.get l a >>= λ b → Lens.put r b c >>= Lens.put l a
  ; get = λ a → Lens.get l a >>= Lens.get r
  ; PutGet = ?
  ; GetPut = ? }

```

The *put* and *get* functions for a composite lens can be pretty straightforwardly constructed from the *put* and *get* functions of the component lenses. As part of the definition of $_ \leftrightarrow _$, we also need to prove that *PutGet* and *GetPut* hold for the pair of *put* and *get* we provide. For *PutGet* we need to show that, for all *a*, *a'* : *A*, and *c* : *C*, from a proof of *put* *a* *c* $\mapsto$ *a'* we can construct a proof of *get* *a'* $\mapsto$ *c*. That is, we need to write a function converting an inhabitant of the type *put* *a* *c* $\mapsto$ *a'* to one of the type *get* *a'* $\mapsto$ *c*. AGDA automatically expands the two types following the definition of $_ \mapsto _$:

$$\begin{aligned}
& \text{put } a \ c \mapsto a' \\
&= (b : B) \times (\text{Lens.get } l \ a \mapsto b) \times \\
&\quad (b' : B) \times (\text{Lens.put } r \ b \ c \mapsto b') \times (\text{Lens.put } l \ a \ b' \mapsto a')
\end{aligned}$$

and

$$\begin{aligned} & \text{get } a' \mapsto c \\ &= (b : B) \times (\text{Lens.get } l \ a' \mapsto b) \times (\text{Lens.get } r \ b \mapsto c) \end{aligned}$$

Thus all we need to do is convert a five-tuple to a three-tuple of the above types. This is easy: Applying `Lens.PutGet l` to the fifth component of type `Lens.put l a b' ↦ a'` we obtain an inhabitant of type `Lens.get l a' b'`, and applying `Lens.PutGet r` to the fourth component of type `Lens.put r b c ↦ b'` we obtain an inhabitant of type `Lens.get r b' c`. Hence the proof term for `PutGet` is simply

$$\lambda (b, x, b', y, z) \rightarrow (b', \text{Lens.PutGet } l \ z, \text{Lens.PutGet } r \ y)$$

With a similar reasoning, we can also construct a proof term for `GetPut`:

$$\lambda (b, x, y) \rightarrow (b, x, b, \text{Lens.GetPut } r \ y, \text{Lens.GetPut } l \ x)$$

completing the definition of `↔`. For the proofs in the rest of this paper we will merely provide a sketch, but these proofs all have their formal counterparts in AGDA as illustrated above, completely guaranteeing their validity.

4 BiGUL and its lens semantics

Various bidirectional programming languages have been built based on the idea of lenses. Such languages would contain a set of primitive lenses, like those we build by *iso-lens*, and a set of combinators which combine smaller lenses into larger ones, like what `↔` does. Usually these languages are designed to suggest that what they describe are *get* programs, but it has recently been argued that the opposite *putback-based* approach should be adopted instead. The argument is based on the following theorem:

Theorem 1 (putback is the essence). *Define, for every $mx, my : \text{Par } A$, a type $mx \approx my$ to mean that $mx \mapsto z$ if and only if $my \mapsto z$ for any $z : A$. Then for any $S, V : \text{Set}$ and two lenses $l, r : \text{Lens } S \ V$ such that*

$$\text{Lens.put } l \ s \ v \approx \text{Lens.put } r \ s \ v \quad \text{for all } s : S \text{ and } v : V,$$

we have

$$\text{Lens.get } l \ s \approx \text{Lens.get } r \ s \quad \text{for all } s : S.$$

The theorem tells us that, when designing a bidirectional combinator, if we can justify the design of its *put* component, then its *get* component needs no further justification, since there is no other *get* that can be paired with the same *put*. This is in sharp contrast with the *get*-based approach, with which the designer needs to choose for each *get* one or a few possible *put* strategies to be paired with the *get*. This complicates the design as there would be multiple combinators with

the same *get* semantics, distinguished only by differences in their *put* semantics — this is rather counterintuitive since the programmer is supposed to think in terms of *get* instead of *put* when programming in *get*-based languages. Also it is harder to justify the design as to why the set of particular *put* strategies are chosen, and it may well happen that, after the programmer describes a *get*, the corresponding *put* synthesised by the language is not what the programmer wants. All these problems simply do not exist for the putback-based approach.

Following the putback-based approach, our language BiGUL describes *put* functions, i.e., updates to a source using a view. In general, an update proceeds by decomposing the source into several parts to be updated independently (Section 4.5), and accordingly rearranging the view to match with these parts (Section 4.6), until the source and the view are reduced to the extent that atomic operations can be applied (Section 4.1). When both the source and the view are lists, there is a powerful aligning operation for synchronising the two lists (Section 4.4). And we can do case analyses on either the source (Section 4.2) or the view (Section 4.3) for describing more flexible update strategies.

In the rest of this section we will present the lens semantics of each of the bidirectional operations mentioned above.

informal representation of deep embedding

Design strategy: start from a natural semantics of *put*, and add checks that guarantee existence of *get*

4.1 Replacing, failure, and skipping

The three most basic lenses are replacing, failure, and skipping. *Replacing* is applicable when the source and the view are of the same type, and its *put* semantics is discarding the original source and returning the view as the updated source. As mentioned in Section 3, we can in fact derive the lens from the identity isomorphism:

$$\begin{aligned} \text{replace-lens} &: \{S : \text{Set}\} \rightarrow \text{Lens } S \ S \\ \text{replace-lens} &= \text{iso-lens } (\text{record } \{ \text{to} = \text{return} ; \text{from} = \text{return} ; \dots \}) \end{aligned}$$

The corresponding *get* semantics is therefore just the identity function. Also mentioned in Section 3 is that we can derive the *failure* lens from the empty isomorphism in the same way:

$$\begin{aligned} \text{fail-lens} &: \{S \ V : \text{Set}\} \rightarrow \text{Lens } S \ V \\ \text{fail-lens} &= \text{iso-lens } (\text{record } \{ \text{to} = \text{fail} ; \text{from} = \text{fail} ; \dots \}) \end{aligned}$$

Both *put* and *get* of this lens fail for any input. On the other hand, *skipping*, whose putback semantics discards the view and returns the original source, is not an instance of *iso-lens*. It is the first lens that we define from scratch.

$$\begin{aligned} \text{skip-lens} &: \{S : \text{Set}\} \rightarrow \text{Lens } S \ \top \\ \text{skip-lens} &= \text{record } \{ \text{put} = \lambda s _ \rightarrow \text{return } s ; \text{get} = \lambda _ \rightarrow \text{return } \text{tt} ; \dots \} \end{aligned}$$

Note that the view type is specified as $\top$, whose inhabitant can only be `tt`: had the view type been a more complex type, there would have been no way for *get* to

```

caseS-lens : (S V : Set) → List (CaseSBranch S V) → Lens S V
caseS-lens S V bs = record
  { put = put-with-adaptation bs s v
    (λ s' → put-with-adaptation bs s' v (λ _ → fail))
  ; get = get ; ... }
where
  branch : {X : Set1} → (Lens S V → X) → ((S → Par S) → X) →
    CaseSBranchType → X
  branch f g (normal l) = f l
  branch f g (adaptive u) = g u
  put-with-adaptation : List CaseSBranch → S → V → (S → Par S) → Par S
  put-with-adaptation [] s v f = fail
  put-with-adaptation ((p , b) :: bs) s v f =
    p s ≫ λ matched →
      if matched then branch (λ lens → Lens.put lens s v ≫ λ s' →
        p s' ≫ λ matched' →
          assert matched' then return s')
        (λ u → u s ≫ f) b
      else put-with-adaptation bs s v f ≫ λ s' →
        p s' ≫ λ matched' → assert not matched' then return s'
  get : List CaseSBranch → S → Par V
  get [] s = fail
  get ((p , b) :: bs) s =
    p s ≫ λ matched →
      if matched then branch (λ lens → Lens.get lens s) (λ _ → fail) b
      else get bs s

```

Fig. 2. Definition of source case analysis

decide what to return. In general, *put* must embed all view information into the updated source so *get* can recover the entire view from the updated source, establishing *PutGet*. $\top$ is a type with no information, and hence *Lens.put skip-lens* can safely ignore its view argument of type $\top$.

4.2 Case analysis on source

The next lens, *caseS-lens*, is for doing case analysis on the source. The basic idea of the putback semantics of *caseS-lens* is to choose from a list of branches the first one that matches the source, and then execute the putback semantics of the lens associated with the chosen branch. It is, however, difficult to find a *get* to pair with this *put*: the obvious *get* which selects the first branch that matches its source argument does not work, because it may well happen that the updated source produced by *put* matches a different branch from the original source, so in general we would have to guarantee *PutGet* for *put* and *get* coming respectively from any two branches, which is unmanageable. A more manageable solution is to insert a dynamic check at the end of *put* to ensure that the updated source

must match the same branch as the original source, but in practice this solution is too restrictive to be useful. We bump right into an obstacle for our very first combinator!

Our solution is to invent a special kind of branches called *adaptive* branches, in addition to *normal* branches. The two kinds of branches are defined by the following datatype:

```
data CaseSBranchType (S V : Set) : Set1 where
  normal  : Lens S V    → CaseSBranchType S V
  adaptive : (S → Par S) → CaseSBranchType S V
```

A normal branch contains a lens, while an adaptive branch contains a source transformation. With each branch we also need to associate a decidable predicate on the source type specifying when a source matches the branch, so the complete definition of the type of branches is

```
CaseSBranch : Set → Set → Set1
CaseSBranch S V = (S → Par Bool) × CaseSBranchType S V
```

The type of *caseS-lens* is then

```
caseS-lens : (S V : Set) → List (CaseSBranch S V) → Lens S V
```

and its definition is shown in Figure 2. We might think of a source case analysis as still consisting mainly of normal branches, but now for *put* we allow the source to match an adaptive branch, which transforms the source such that it matches a normal branch. More explicitly, the case analysis may be run twice when executing *put*: if a normal branch is chosen the first time, then the case analysis succeeds, and we execute the *put* associated with the branch; otherwise, we adapt the source and rerun the case analysis, and must match a normal branch this time. After the *put* of a normal branch is executed, we must ensure that the updated source satisfies the predicate of the branch and also none of the predicates of the previous branches, so a subsequent execution of *get* on the updated source would choose the same branch as *put*, establishing *PutGet*.

4.3 Case analysis on view

For case analysis on the view, there is still a list of branches, with each of which a lens is associated — there are no adaptive branches this time. We must be more careful when it comes to the view, though: As we mentioned in Section 4.1, view information must be carefully managed and completely embedded into the source. When a branch matches the view, the amount of view information that remains to be embedded into the source can actually be reduced. For example, if there is a branch that matches an integer view which equals zero, then for this branch we may even perform no update on the source as long as we can guarantee that the subsequent *get* will choose this branch, in which case *get* can simply return zero — the information that the view is zero is embedded implicitly in the


```

caseV-lens : (S V : Set) → DecEq V → List CaseVBranch S V → Lens S V
caseV-lens S V dec bs = record { put = put ; get = get ; ... }
where
  guarded-return : S → V → V → Par S
  guarded-return s' v v' with dec v v'
  guarded-return s' v v' | inj1 eq = return s'
  guarded-return s' v v' | inj2 neq = fail

  put : List CaseVBranch S V → S → V → Par S
  put [] s v = fail
  put ((_, lens, iso) :: bs) s v = catch (Iso.from iso v) (Lens.put lens s)
                                     (put bs s v >>= λ s' →
                                     catch (Lens.get lens s' >>= Iso.to iso)
                                     (guarded-return s' v) (return s'))

  get : List CaseVBranch S V → S → Par V
  get [] s = fail
  get ((_, lens, iso) :: bs) s = catch (Lens.get lens s >>= Iso.to iso) return
                                     (get bs s >>= λ v →
                                     catch (Iso.from iso v) (λ _ → fail) (return v))

```

Fig. 3. Definition of view case analysis

control flow rather than explicitly in the updated source. Thus, in the definition of `CaseVBranch` below, we use a partial isomorphism on the view type which converts it to a less informative type, instead of just a decidable predicate as in `CaseSBranch`:

```

CaseVBranch : Set → Set → Set1
CaseVBranch S V = (V' : Set) × Lens S V' × Iso V' V

```

For the comparison-with-zero example above, we would set V' to be $\top$, indicating that there is no more information to be embedded into the view after a successful matching, and set the *to* and *from* components of the isomorphism to $\lambda _ \rightarrow \text{return } 0$ and $\lambda n \rightarrow \text{assert } n == 0 \text{ then return tt}$. A branch is considered matched with a view when the *from* conversion of the associated isomorphism succeeds for the view.

We can now define *caseV-lens*, whose type is

```

caseV-lens : (S V : Set) → DecEq V → List CaseVBranch → Lens S V

```

and its definition is shown in Figure 3. (We will come to the `DecEq` requirement shortly.) Its *put* tries to match the view with each branch, and proceed to execute the associated putback semantics when a branch succeeds. For *get*, however, there is no obvious way to decide which branch to go, as we are not given a view to start with like *put*. It seems that we can only try to execute each of the branches until we reach one that computes a view successfully. This presents a serious problem, as we cannot easily guarantee that executing *put* followed by *get* would use the same branch. Our compromised solution is to make *put* do

an expensive check after a branch is matched and executed, requiring that the *get* for each of the previous branches must either fail or produce the same view as the input one (i.e., verifying *PutGet* directly). The `DecEq V` argument of *caseV-lens*, which expands to $(v \ v' : V) \rightarrow (v \equiv v') \uplus \neg (v \equiv v')$ and says that *V* is equipped with a decidable equality, is needed for this check.

4.4 List alignment

4.5 Source update

A fundamental operation is performing updates to parts of the source. We follow the approach taken by FLUX [1] and require that updates must be independent — that is, the same part of a source cannot be updated more than once — to avoid the problem of sequencing updates. It turns out that a pattern matching notation suits this task perfectly: pattern matching is arguably the most intuitive way to deconstruct a source, and we can specify apparently independent updates by writing them at the variable positions in a pattern.

Since AGDA is fully dependently typed, we can naturally define patterns in a type-directed way such that nonsensical patterns are syntactically ruled out. In a dependently typed language, this task starts from the construction of a *universe*, i.e., a datatype whose inhabitants are interpreted as types. In this paper we present a simplified universe to make the presentation easier to follow, but the actual AGDA implementation uses a more powerful universe capable of expressing mutually inductive datatypes.

```
data U : Set1 where
  var  : U
  k    : (A : Set) → DecEq A → U
  one  : U
  _⊕_  : U → U → U
  _⊗_  : U → U → U
  list : U → U → U
```

4.6 View rearrangement

5 Examples

bookstore for align, update, and rearrangement

BrYACC for source and view cases (probably not)

source/view if, map in terms of align

6 Discussion

recursion (the lack thereof)

extensions: for example, iteration, view dependency

design/programming phase: design phase needs to consider *put* and *get* together, but programming phase is about *put* only

Haskell implementation

Towards a dependently typed version

stems from work on BiFLUX; comparison with putlenses

Theory construction in terms of AGDA programming

Regarding formalisation: “[W]e are interested in extending what can be calculated precisely because we are not interested in the calculations themselves. Developing techniques that allow more of a derivation to be calculated allows attention to be focussed on the creative parts, which cannot be calculated.” [3]

[7]

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Appendix: AGDA syntax

An AGDA datatype is defined by listing all of its constructors along with their types in full. A datatype can be *parametrised*, as in the case of `Maybe`, which has a parameter $A : \text{Set}$, specified between the name of the datatype and the colon on the first line. (`Set` is the type of all (small) types.) Parameters can be referred to throughout a datatype definition.

Here the datatype `Par` is *indexed* by `Set` (which appears to the right of the colon on the first line), meaning that `Par` is in fact a `Set`-indexed family of simultaneously defined types.

As for the constructors, let us look at the simplest case, `fail`: The constructor `fail` constructs a value of type `Par A` for any $A : \text{Set}$. There is thus one argument to `fail`, namely $A : \text{Set}$, and the type of `fail`'s result depends on the value of this argument (which happens to be a type). This is an example of *dependent function types*, whose result type can depend on the value of the argument. Logically, dependent function types are read as universally quantified propositions — the dependent function type $\{x : A\} \rightarrow B$ is read as “for all $x : A$ the property B holds for x ”. (Ordinary function types, on the other hand, denote implications logically.)

Back to `fail`: most of the time AGDA can infer what `fail`'s argument should be based on how `fail` is used, so we make the argument *implicit* by enclosing it in curly brackets. (Explicit arguments, on the other hand, are enclosed in round brackets.) We can then invoke `fail` without supplying the argument, leaving it for AGDA to infer. We de-emphasise implicit arguments typographically when they cause visual annoyance, as in this case.

An AGDA name can be used as an infix (or, in general, mixfix) operator when the name contains underscores, in which case the underscores indicate where the arguments should go. For example, we can apply `_>>=_` to arguments mx and f by writing either `_>>= mx f` or `mx >>= f`.

The type $\top : \text{Set}$ has exactly one constructor `tt` : $\top$, while the type $\perp : \text{Set}$ has no inhabitant — logically they are read as truth and falsity respectively. The type family `_≡_` : $\{A : \text{Set}\} \rightarrow A \rightarrow A \rightarrow \text{Set}$ is defined such that $x \equiv y$ is inhabited if and only if x and y are equal. The type former `_⊔_` : $\text{Set} \rightarrow \text{Set} \rightarrow \text{Set}$ is disjoint union (disjunction) whose constructors are `inj1` : $\{A B : \text{Set}\} \rightarrow A \rightarrow A \uplus B$ and `inj2` : $\{A B : \text{Set}\} \rightarrow B \rightarrow A \uplus B$, and the type former `_×_` : $\text{Set} \rightarrow \text{Set} \rightarrow \text{Set}$ is cartesian product (conjunction), whose constructor is `_,_`. The symbol ‘ $\times$ ’ is also overloaded to denote *dependent pair types* — the inhabitants of a dependent pair type $(x : A) \times B$ are pairs where the type of the second component depends on the value of the first component. Logically, dependent pair types are read as existentially quantified propositions — $(x : A) \times B$ is read as “there exists $x : A$ such that the property B holds for x ”.

An underscore in a right-hand side expression instructs AGDA to infer what should be placed there.

An AGDA record is defined by listing the names and types of all its components, and a component type can depend on previous components. To access a component of an inhabitant of a record type, we should use the corresponding

component extraction function. For example, $\text{Lens.get} : \{S \ V : \text{Set}\} \rightarrow \text{Lens } S \ V \rightarrow S \rightarrow \text{Par } V$ is for extracting the *get* component of a **Lens**.